CSC 255 Objects and Algorithms

Group Project Tips for Success

Lay the Groundwork 📝

Before writing a single line of code, your group needs to get organized. This initial investment of time will prevent major headaches later.

- **Establish Communication:** Choose one primary place for all project-related communication. A dedicated **Slack** or **Discord** channel is perfect for this. It keeps discussions organized and out of personal texts or emails.
- **Define Roles & Responsibilities:** While everyone will code, designate leads for different areas. For example, who is the point person for the database? Who is responsible for the final presentation? Who will manage the task board? This creates ownership.
- **Choose Your Tools:** Collectively agree on the technology stack, IDEs, and any libraries you'll use. Make sure everyone has the same development environment set up and working from day one.

Master Version Control with Git 🐙

Using version control is not optional; it's essential for collaborative coding.

- **Use a Central Repository:** Create a project on **GitHub**, GitLab, or a similar platform. This will be the single source of truth for all your code.
- **Embrace Branching:** Never commit directly to the main branch. Adopt a branching strategy. A common one is the **feature branch workflow**:
    1. Create a new branch for every new feature or task (e.g., login-page, fix-database-bug).
    2. Do all your work on that branch.
    3. When the feature is complete, submit a **pull request** (PR) to merge it into the main branch.

- **Review Each Other's Code:** A pull request is an opportunity for at least one other team member to review the code before it's merged. This catches bugs, improves code quality, and helps everyone learn the entire codebase.

Communicate Constantly and Clearly 🗣️

Technical problems are often communication problems in disguise.

- **Hold Regular Check-ins:** Schedule short, daily or bi-daily meetings (often called "stand-ups"). Each person should briefly state what they did yesterday, what they plan to do today, and if they are blocked by anything.
- **Be Honest and Proactive:** If you're stuck, say so immediately. It's better to ask for help early than to struggle for days and fall behind. Similarly, if you're running ahead, ask if anyone needs a hand.
- **Document Decisions:** Keep a simple log of important decisions made during meetings so everyone can refer back to them.

Standardize Your Code ✨

A consistent codebase is easier to read, debug, and maintain. It should look like it was written by one person, not four different people.

- **Agree on a Style Guide:** Decide on conventions for naming variables (camelCase vs. snake_case), indentation (tabs vs. spaces), and commenting.
- **Use Linters and Formatters:** Tools like Prettier, ESLint, or Black can automatically format the code to match the agreed-upon style. This eliminates tedious arguments about formatting.

Break It Down and Track Everything ✅

A big project can feel overwhelming. The key is to break it into small, manageable pieces.

- **Decompose the Project:** Break the entire project down into the smallest possible tasks or "issues." A good task is one that can be completed in a few hours to a day.
- **Use a Task Board:** Use a simple Kanban-style board like **Trello**, **Asana**, or **GitHub Projects** to visualize your workflow. Create columns like "To-Do," "In Progress," and

"Done." Assign each task to a person so it's clear who is working on what. This makes progress visible to everyone and holds members accountable.

Integrate and Test Often 🔄

Avoid the nightmare of a last-minute integration where nothing works together.

- **Merge Frequently:** Don't let branches live for weeks. Merge small, completed features back into the main branch as often as possible (at least daily). This practice, known as **continuous integration**, prevents massive, complex merge conflicts.
- **Test Your Code:** Write tests for your features. This ensures that when someone else adds new code, they don't accidentally break your part of the project.